AYUSH PANDEY

Lead AI Scientist | ML Engineer | Computer Vision

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PROFESSIONAL SUMMARY

AI Scientist and ML Engineer with 5+ years of experience spanning clinical AI, computer vision, and enterprise data science. Currently Lead AI Scientist at Scanvue AI, building end-to-end deep learning systems for medical imaging (TB detection) and animal biometrics (bovine ID). Previously at Tata Consultancy Services and IIT Madras HTIC. Expertise in CNNs, Vision Transformers, YOLO, Attention U-Nets, and full ML pipeline engineering from data to deployment.

PROFESSIONAL EXPERIENCE

Scanvue AI CURRENT — Remote

Lead AI Scientist — May 2025 – Present

- Designed and shipped a **3-phase clinical TB detection pipeline** on chest X-rays — lung segmentation (YOLOv8), lesion segmentation (Attention U-Net, **Dice 0.830**), and ROI classification (ViT-Small, **AUC 0.988**) — deployed as ScanView v3.0.4 with DICOM support and auto-generated PDF clinical reports.
- Unified **~11K chest X-ray images** from Roboflow, SFLTB, and TBX11K; addressed severe class imbalance via weighted sampling and focal loss, improving lesion recall significantly.
- Architecting an end-to-end **bovine biometric ID system** using muzzle-pattern recognition for non-invasive, tag-free livestock tracking; Phase 1 (instance segmentation pipeline) **shipped May 2026**; Phase 2 (pattern matching & re-ID) in active development.
- Owns the full ML lifecycle: data strategy, annotation workflow, model training, evaluation, and deployment-ready FastAPI inference.

Tata Consultancy Services — New Delhi, India

Data Analyst & ML Engineer — December 2021 – May 2025

- Designed SQL/HiveQL/ImpalaQL queries for client data extraction across Oracle and distributed Hive/Impala systems.
- Built ensemble and boosting classification models (delinquency prediction) achieving **90% test accuracy**.
- Developed Power BI customer analytics dashboards; identified optimal offer strategies driving **20% increase in customer acquisition**.
- Designed Hive-based fraud detection database system aggregating failed transaction data from multiple business sources.
- Engineered Python-based ETL pipelines consuming REST APIs with business-rule filtering, reducing data errors by **15%**.

Healthcare Technology Innovation Centre — IIT Madras — Chennai, India

Algorithm Developer & Data Science Intern — July 2021 – October 2021

- Built an ECG-based sleep stage classification model using RNNs and signal processing achieving **87% test accuracy**.

- Developed mathematical models identifying biometrics influenced by exercise; findings drove core feature redesign of the Repose health app.
- Contributed to ML feature development, supporting a **10% increase in app-purchase revenue**.

RESEARCH PROJECTS & COMPETITIVE ML

TB Detection System — ScanView v3.0.4 *PyTorch · YOLOv8 · Attention U-Net · EfficientNet-B4 · ViT-Small · TensorFlow · DICOM*

- Built a multi-phase clinical chest X-ray screening pipeline for Tuberculosis detection; lesion segmentation achieved **Dice 0.830**, ROI classification reached **AUC 0.988**.
- Unified **~11K images** from Roboflow, SFLTB, and TBX11K; resolved severe class imbalance via weighted sampling and focal loss.
- Deployed as ScanView v3.0.4 with DICOM input support, heatmap visualisation, and auto-generated clinical PDF reports.

Dog Muzzle Biometric Re-ID System *ResNet-18 · ArcFace · CBAM · FPN · Triplet Loss · SAM · PyTorch*

- Built dog identity re-identification system using metric learning (triplet loss, ArcFace) with CBAM attention and FPN multi-scale features; full-face crops achieved **96.8% Top-5 accuracy** vs. 34.67% for muzzle-only inputs.
- Developed Laplacian-based frame sharpness selection for video input preprocessing. Invention Disclosure Form in preparation for IP filing.

AMIA Public Challenge 2026 — VinDr-CXR *YOLOv8x · A100 GPU · 14-class pathology detection*

- Trained YOLOv8x on VinDr-CXR for 14-class chest pathology detection; best **mAP50 ~0.379** at epoch 26 on NVIDIA A100.

Harmful Brain Activity Classification *VGG16 · ResNet-3D · EEG · Kaggle / Harvard Medical School*

- Classified brain seizure patterns from EEG graphs (Harvard Medical School via Kaggle); VGG16 + ResNet-3D achieving **KL Divergence loss 0.45**.

VSSUT Satellite Launch Vehicle (VSLV) *Telemetry · IMU · Python · Avionics*

- Telemetry Lead and Head of AI/ML on India's first student-built reusable rocket; built thrust prediction system improving fuel/motor quality by **45%** and real-time trajectory tracking system improving operational efficiency by **30%**.

TECHNICAL SKILLS

Languages	Python, SQL, HiveQL, ImpalaQL
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-Learn
Computer Vision	YOLOv8, U-Net, Attention U-Net, ViT, ResNet, EfficientNet, VGG, CBAM, FPN, ArcFace, DEiT
LLMs & GenAI	RAG, Ollama, LLaMA, VLMs, Hugging Face, LangChain, LangGraph, LlamaIndex
MLOps & Deployment	MLflow, Docker, FastAPI, Streamlit, Weights & Biases
Vector Databases	Pinecone, ChromaDB, FAISS
Data & Visualization	Power BI, Pandas, NumPy, Matplotlib, Seaborn

Platforms	Kaggle, DrivenData, Google Colab, AWS (basic)

EDUCATION

Veer Surendra Sai University of Technology (VSSUT) Burla, Odisha

B.Tech — Chemical Engineering July 2017 – July 2021

CGPA: 7.29 | First Class Distinction

CERTIFICATIONS

- Computational Thinking using Python — MITx on edX
- Machine Learning — Stanford Online (Andrew Ng)
- Combinatorics and Probability — UC San Diego on Coursera

ACHIEVEMENTS & AWARDS

- 1st Runner-up — Tata Crucible Hackathon 2020
- 1st Runner-up — Smart India Hackathon 2020
- Best Research Project — Sristi Summer School 2019
- Active Kaggle competitor — TB detection, BirdCLEF 2026, VinDr-CXR (AMIA 2026), Harmful Brain Activity Classification
- Invention Disclosure Form in progress for dog biometric re-identification system